

SIDDHANT K. SANCHETI

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SKILLS

Languages: Python, Java, Ruby, Bash, TypeScript, JavaScript, SQL, HTML5, CSS3, Shell.
Technologies/Tools: AWS, Google Cloud Platform, Kubernetes, Docker, CI/CD, Jenkins, Git, Agile, RESTful APIs, RPC, Bash, Load Balancing.
Frameworks: AWS CDK, React, Node.js, Express.js, Flask, Django, Apache Spark, Bootstrap, Springboot, Kafka.
Databases: AWS RDS, DynamoDB, PostgreSQL, MongoDB, Firebase.
AI/ML Stack: TensorFlow, PyTorch, JAX, LangChain, VertexAI, FAISS, Scikit-learn, Computer Vision, OpenCV, Pandas, NumPy, TensorRT, Keras, MCP, Sagemaker, LLMs, RAG, Natural Language Processing.

WORK EXPERIENCE

Software Development Engineer, Amazon Web Services, Seattle, WA May 2025 - Present
RDS SQL to IAM auth migration (Primary Skill: CDK(IaC), Java, CloudWatch, Lambda, TypeScript, RDS, SQL, Ruby)

- Led **zero**-downtime migration of Aurora PostgreSQL authentication from static credentials to IAM tokens across **110+** Lambda functions and CLI tools in **17** regions, handling **150-190** connections/sec and achieving **100%** IAM adoption with zero password fallback.
- Designed **CDK** infrastructure with least-privilege IAM policies and resolved critical production issues including **S3** client connection leaks, file descriptor exhaustion, and cross-region token signing, improving system stability for high-throughput multi-region database workloads.

AI Agent Orchestration for code automation (Primary Skill: Python, CDK, Go, DAG, PostgreSQL)
Co-designed and built a **Python**-based multi-agent code automation system that parsed SIM task tables into dependency **DAGs**, scheduled **5-10** parallel agent workflows, and added persistent state, resumable execution, and live monitoring, reducing **10**-task implementation cycles across a **195+** package ecosystem from **5-7 days** to **1-2 hours**.

Fleet Capacity Offset Management System (Primary Skill: Java, DynamoDB, Coral RPC, CLI Tooling, Shell)
Built a self-service fleet capacity management system using **Java**, **DynamoDB**, **RPC APIs**, and safe-by-default CLIs, reducing scaling workflows from **30 mins** to **2-4 mins** across **5,000+ EC2** hosts while preventing conflicting updates and error-prone manual production changes.

Software Engineer, GrantAide, San Jose, USA Sept 2024 - Jan 2025
AI-Driven Grant Writing Platform (Primary Skill: React, TypeScript, GPT-4, Flask, AWS, Vertex AI, LangChain, FAISS)

- Built and scaled an AI-driven grant writing platform using **React**, **GPT-4**, **Flask**, **AWS (S3, Amplify, EC2, Elastic Beanstalk)** and **Stripe APIs** improving platform responsiveness **5x** and strengthening payments, subscriptions, and overall user experience.
- Engineered responsive frontend and **RAG** workflows with **Material-UI**, **Firebase**, **Vertex AI**, **LangChain**, and **FAISS**, reducing frontend load times by **35%**, cutting query latency from **200ms** to **50ms**, improving response accuracy by **80%**, and increasing grant application success rates by **85%**.

Software Developer - Full Stack, Forsk Technologies, Jaipur, India Mar 2021 - Jun 2022
Campus Portal Modernization (Primary Skill: React.js, Node.js, Express.js, GraphQL, AWS, Jenkins, Docker)

- Architected an AI-enabled Campus Portal using **React**, **Node.js**, **Express.js**, **AWS Lambda**, **DynamoDB**, and **ECS**, integrating **TensorFlow / PyTorch**-based chatbot and feedback intelligence, reducing onboarding time by **50%** for **5,000+ MAU** and improving service adoption by **35%**.
- Optimized data access and user workflows with **GraphQL**, reducing data load times by **20%** and improving frontend performance by **30%**, which increased portal usability and engagement.
- Automated deployment and service operations with **Jenkins**, **GitHub Actions**, **Docker**, and **Grafana**, increasing release frequency by **40%** and maintaining **99.9%** uptime in production.

Software Engineer, Pantech Prolabs, Pune, India Jun 2020 - Feb 2021
Distributed Job Platform Data Processing (Primary Skill: Python, Kafka, Apache Spark, Django, Flask)

- Optimized asynchronous data pipelines for a job listing platform using **Python**, **Kafka**, **Zookeeper**, and **Apache Spark**, reducing processing time by **30%** and enabling scalable **ETL** workflows for high-volume data handling.
- Enforced secure **RESTful APIs** with **Django**, **Flask**, **OAuth2**, **JWT**, and **Terraform**, protecting data for **100,000+** users, reducing unauthorized access incidents by **45%**, and cutting manual provisioning effort by **35%** through automated, scalable deployment workflows.

PROJECTS

Advanced Lane Detection using Computer Vision and Deep Learning Techniques Aug 2023 - May 2024
Devised a lane detection system for autonomous vehicles using **CARLA** simulation, **ENet**, **Hourglass** neural networks, and **RANSAC** algorithms, achieving **93.19%** classification accuracy and **3.89%** MAE, ensuring reliable detection under varied conditions.

Smart Image Store Sept 2023 - Dec 2023
Created an Image store where users can upload and retrieve images based on object tags extracted using **Google Cloud Vision** API, optimized search with a HashMap-based algorithm, and utilized **MongoDB**, **React.js**, **load balancing**, and auto-scaling in **AWS**.

EDUCATION

San Jose State University, San Jose, USA **GPA: 3.81**
Masters of Science in Software Engineering Aug 2022 - May 2024
Savitribai Phule Pune University, Pune, India **GPA: 3.81**
Bachelors of Engineering Aug 2016 - May 2020